BiSAM-CD: Zero-Shot Remote Sensing Change Detection via Bidirectional Temporal Memory in SAM2

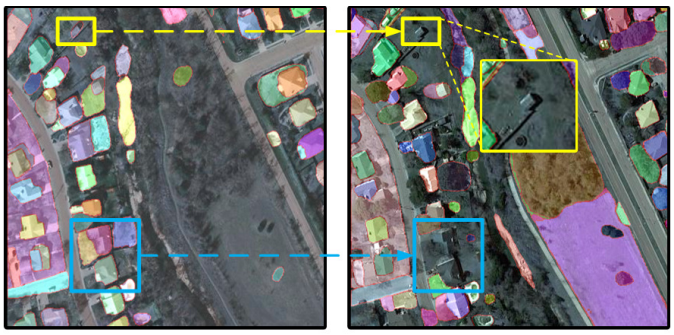
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***Abstract*—The reliance on annotated training data remains a critical bottleneck for remote sensing change detection, particularly in scenarios requiring rapid response. To address this challenge, our framework operates in a fully training-free paradigm that leverages the zero-shot (ZS) generalization capability and temporal memory mechanism of the segment anything model 2 (SAM2). By redefining change detection as a bidirectional object tracking task, our method constructs ascending and descending frame sequences with adaptive interpolation to mitigate environmental noise. Two lightweight modules are introduced: Module A suppresses pseudo-changes through cross-sequence mask validation, while Module B stabilizes tracking via nonlinear frame interpolation. Evaluated on WHU-CD, LEVIR-CD, and SECOND datasets, our approach achieves $F1$ scores of 85.31% (mask prompts), 57.43% (SAMPoly prompts), and 39.35% (auto-masks), respectively, rivaling supervised models like FC-Siam-Cone (89.53%) in building change detection while demonstrating ZS generalization to multiclass environments (SECOND dataset) without requiring semantic transition modeling. Ablation studies confirm the necessity of each module [temporal object tracking consistency and cross-sequence change verification (TOTC-CSCV) and bidirectional frame sequence construction with adaptive interpolation (BFSCAI)], with combined contributions improving $F1$ by 27.17% over the baseline. The framework processes 1024 × 1024 images in 0.0325 s (Base+) with <1 GB memory, achieving 4× speedup over supervised counterparts. The framework demonstrates practical potential in resource-constrained environments, showing particular promise in rapid disaster assessment (WHU-CD) and multiclass land cover monitoring (SECOND). Its dependency on external prompts is mitigated by compatibility with diverse sources (OWLv2, SAMPoly, and SAM2 auto-masks), though autonomous prompt generation remains future work. The code will be open at: https://github.com/KimotaQY/BiSAM-CD**

***Index Terms*—Bidirectional temporal consistency, pseudo-change suppression, remote sensing change detection, segment anything model 2 (SAM2), training-free framework, zero-shot (ZS) learning.**

I. Introduction

CHANGE detection in remote sensing imagery plays a vital role in monitoring environmental dynamics, urban expansion, and disaster assessment. Traditional approaches rely heavily on pixel-level or object-level annotations to train deep learning models, achieving remarkable performance in controlled scenarios. However, the acquisition of high-quality labeled data remains labor-intensive and time-consuming, especially for large-scale satellite imagery with complex geospatial variations. This limitation becomes even more pronounced in real-world applications requiring rapid response, such as postdisaster damage assessment or illegal land use detection, where timely analysis is critical but labeled historical data may be unavailable.



(a) (b)

Fig. 1. Segmentation results of SAM on bi-temporal images. (a) Segmentation result of the first temporal phase. (b) Segmentation result of the second temporal phase.

Recent advances in semi-supervised (SS) [1], [2] and unsupervised [3], [4], [5] change detection methods have partially alleviated the dependency on labeled data. However, these approaches still require extensive model fine-tuning on specific datasets, which introduces additional computational overhead and limits their generalization across diverse geographical regions. Meanwhile, the emergence of foundational vision models [6] has revolutionized generic image segmentation tasks with their zero-shot (ZS) capabilities. Despite their potential, directly applying these models to remote sensing change detection faces critical challenges (shown in Fig. 1): 1) temporal inconsistency caused by illumination variations and seasonal changes often leads to false alarms when naively comparing per-frame segmentation results (blue boxes) and 2) high-resolution satellite images with small objects exacerbate the difficulty of accurate target tracking and mask alignment (yellow boxes).

To address these challenges, we propose BiSAM-CD, a training-free framework that redefines change detection as a bidirectional object tracking problem, leveraging SAM2's [7]

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inherent temporal memory mechanism. Unlike conventional methods that focus on architectural improvements or data-hungry training, our key insight lies in exploiting SAM2's ZS generalization to bridge temporal gaps between bi-temporal images. Specifically, we construct ascending and descending frame sequences with adaptive nonlinear interpolation, enabling SAM2 to robustly track objects while mitigating pseudo-changes caused by environmental noise. To further enhance robustness, BiSAM-CD introduces two novel mechanisms: temporal object tracking consistency and cross-sequence change verification (TOTC-CSCV) suppresses fragmented edges by validating mask consistency across sequences, while bidirectional frame sequence construction with adaptive interpolation (BFSCAI) stabilizes tracking through smooth frame transitions. Crucially, our framework operates in a fully training-free paradigm—it requires neither annotated change masks nor backpropagation-based adaptation. Initial object prompts (boxes/masks) can be flexibly sourced from off-the-shelf detectors [8], [9], [10], [11] or user inputs, ensuring immediate deployability in emergency scenarios.

The main contributions of this work are summarized as follows.

Training-Free Paradigm: We propose the first ZS framework for remote sensing change detection that fully utilizes SAM2's generalization power, eliminating the need for task-specific training.

Bidirectional Temporal Consistency: A dual-sequence strategy (ascending/descending) enhanced by adaptive interpolation is designed to handle illumination and seasonal discrepancies, reducing false alarms by 27.17% compared to baseline.

Lightweight Optimization: TOTC-CSCV and BFSCAI synergistically address key challenges in pseudo-change suppression and temporal tracking stability, achieving competitive accuracy ($F1 = 85.31\%$) on the WHU-CD dataset.

Practical Validation: Extensive experiments demonstrate that our method rivals supervised models while requiring zero training time, offering a scalable solution for rapid environmental monitoring.

The remainder of this article is organized as follows. Section II reviews related works. Section III details the proposed framework. Section IV presents experimental results. Section V discusses the broader implications, limitations, and potential applications of our approach. Finally, Section VI concludes the article.

II. Related Work

A. Supervised and Weakly Supervised Change Detection

Traditional change detection methods heavily rely on supervised learning, where architectures like Siamese networks [12], [13], [14], [15] and attention-based fusion modules [16], [17], [18], [19] have achieved remarkable accuracy by comparing multitemporal features. For instance, [20] proposed a dual-stream CNN to capture multiscale building changes, while weakly supervised approaches [21], [22], [23] reduced annotation costs using image-level labels. However, these methods require extensive fine-tuning on domain-specific data, limiting their adaptability to unseen regions or sensors. Furthermore, the computational cost of training complex models hinders real-time deployment in urgent scenarios like disaster response. Recent SS methods [24] partially alleviate annotation dependency but still demand iterative training, failing to achieve true ZS capability. Even the emerging of generative foundation models [25], which synthesizes change data to alleviate annotation burdens, inherently requires large-scale self-supervised pretraining and data generation.

B. Foundational Vision Models in Remote Sensing

The segment anything model (SAM) [6] has revolutionized image segmentation with its ZS generalization, prompting adaptations to remote sensing tasks. Initial efforts like RS-SAM [26] enhanced small-object segmentation through hybrid prompting, while follow-up works explored its potential for specific tasks like building extraction. Concurrently, several studies have investigated adapting SAM for change detection. For instance, [27] fine-tunes FastSAM with trainable adapter modules for limited-data scenarios, and [28] develops a large-scale pretrained foundation model unifying building extraction and change detection. However, a common limitation of these approaches is their continued reliance on learning temporal relationships from annotated data. SAM2 [7], with its video object tracking capability, remains underutilized in multitemporal remote sensing, as its inherent memory mechanism has not been leveraged to address challenges like seasonal variations or illumination shifts.

C. Temporal Object Tracking for Change Analysis

Object tracking techniques, such as optical flow [29], [30], [31], [32] and recurrent networks [33], [34], [35], have been integrated with change detection, primarily in video surveillance, where temporal consistency is high. A recent, innovative work by [36] extends this concept by repurposing a video tracking model for ZS scene change detection, demonstrating the promise of leveraging tracking paradigms without training. However, their method is designed for near-consecutive video frames and does not address the severe appearance variations inherent in long-term bi-temporal remote sensing imagery.

In satellite-based monitoring, [37] combined tracking-by-detection with change analysis but relied on supervised training, which struggles with long-term pairs. Unsupervised tracking frameworks [38], [39], [40], [41] bypass annotations but often yield fragmented masks and false alarms in high-resolution scenes. The absence of a training-free, memory-augmented tracking strategy for satellite time series remains a critical gap.

D. High-Resolution Remote Sensing Processing

High-resolution images pose unique challenges due to dense small objects and large scene coverage. Sliding window strategies [42], [43] and SAHI-based slicing [44] mitigate memory constraints but generate redundant proposals. Recent work [45] introduced topology-aware filtering to refine detection boxes,

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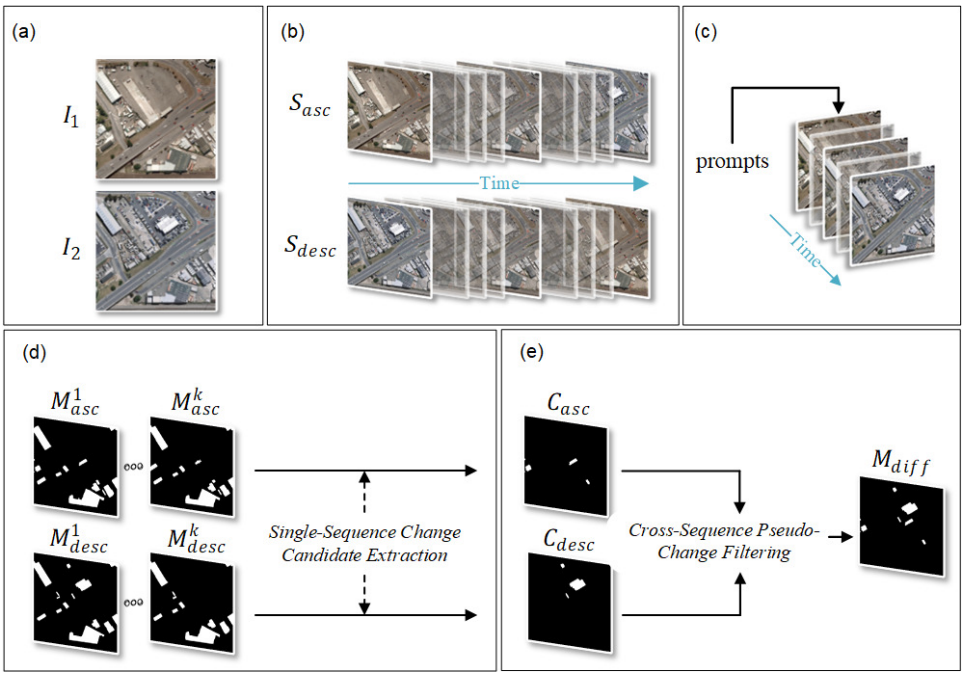


Fig. 2. Framework overview of BiSAM-CD. (a) Input bi-temporal images I_1 and I_2. (b) Bidirectional sequence generation (S_{asc} and S_{desc}) via adaptive interpolation. (c) SAM2-based object tracking initialized with prompts. (d) Change candidate extraction from single-sequence ID continuity checks. (e) Cross-sequence pseudo-change filtering through IoU validation, yielding final mask M_{diff}.

yet manual threshold tuning and computational overhead persist. While SAM2 shows promise in segmenting fine-grained objects, its direct application to large-scale scenes without optimized prompting remains impractical, as demonstrated by the low recall of raw SAM2 outputs in our baseline experiments.

Existing approaches face a fundamental tradeoff: supervised methods achieve high accuracy but require costly annotations, while unsupervised techniques sacrifice precision for flexibility. Foundational models like SAM2 offer ZS potential, but their application to multitemporal analysis is hindered by inadequate exploitation of temporal memory and prompt engineering. Our work addresses these limitations by: 1) redefining change detection as a bidirectional object tracking task; 2) designing lightweight modules to suppress pseudo-changes without training; and 3) optimizing SAM2's temporal consistency for satellite time series. This paradigm shift aligns with the growing demand for scalable, annotation-free solutions in large-scale environmental monitoring.

III. Methodology

A. Overview

Our framework transforms bi-temporal remote sensing images (I_1, I_2) into change detection masks (M_{diff}) through three key stages (see Fig. 2).

Bidirectional Sequence Generation: BFSCAI constructs ascending (S_{asc}) and descending (S_{desc}) frame sequences by interpolating between I_1 and I_2, mitigating abrupt appearance variations.

Memory-Augmented Object Tracking: SAM2 processes S_{asc} and S_{desc} with initial object prompts (boxes from I_1 or I_2), propagating masks and IDs across frames. Only the first frame in each sequence requires initial prompts, while subsequent frames rely on SAM2's memory mechanism.

Consistency-Based Change Detection: Temporal object tracking consistency and cross-sequence change verification (TOTC-CSCV) extracts candidate changes $(C_{\text{asc}}, C_{\text{desc}})$ by comparing head-tail frame ID continuity within each sequence, then filters pseudo-changes via cross-sequence IoU validation.

The entire pipeline operates without model training, leveraging SAM2's ZS tracking capability to achieve label-free change detection.

B. Bidirectional Frame Sequence Construction With Adaptive Interpolation

Directly tracking objects between I_1 and I_2 often fails due to drastic appearance variations (e.g., seasonal changes and shadows). Existing interpolation methods linearly blend pixel values but ignore nonlinear photometric distortions. To address this, we propose an adaptive interpolation strategy that.

1) *Smooth Transition:* Gradually morph I_1 to I_2 to stabilize SAM2's tracking.
2) *Nonlinear Perturbation Modeling:* Simulate real-world lighting/seasonal fluctuations.

Let $I_1, I_2 \in \mathbb{R}^{H\times W\times 3}$ be bi-temporal images. For the ascending sequence $S_{\text{asc}} = \{F^{(0)}, \ldots, F^{(N+1)}\}$

$$F^{(k)} = \underbrace{\alpha_k I_1 + (1-\alpha_k) I_2}_{\text{Linear Blend}} + \underbrace{\beta \cdot \sin\left(\frac{\pi k}{N}\right) \cdot \Delta I}_{\text{Nonlinear Perturbation}} \tag{1}$$

where $\alpha_k = 1 - (k/N + 1)$ controls blending weights, β scales perturbation intensity, and ΔI is a random noise field

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simulating illumination changes. The descending sequence S_{desc} reverses the blending order ($I_2 \rightarrow I_1$).

Bidirectional tracking mitigates unilateral failures.

1) *Forward Tracking (S_{asc}):* Detects disappearances (e.g., demolished buildings).
2) *Backward Tracking (S_{desc}):* Detects appearances (e.g., new constructions).
3) *Conflict Resolution:* If an object is tracked in S_{asc} but lost in S_{desc}, it is flagged as a potential change candidate for further validation.

C. Temporal Object Tracking Consistency and Cross-Sequence Change Verification

1) Theoretical Foundation of Temporal Consistency Constraints: SAM2's temporal tracking mechanism assigns a unique ID to each object and updates its mask across the sequence. We propose the following hypothesis.

Intrasequence ID Consistency: If an object remains unchanged temporally, its ID should persist between the designated first frame (reference frame) and the last frame (ending frame) of the sequence, and its mask shape should exhibit high similarity. Conversely, if the ID is lost in the last frame or the mask similarity drops significantly, the object is considered changed (Fig. 3).

Mathematical Formalization: For a sequence S, let $O_{\text{ref}} = \{o_1, o_2, \ldots, o_n\}$ denote the set of objects in the designated first frame and O_{end} denote the set in the last frame. The changed regions are defined as

$$\text{Changed Regions} = \bigcup o_i \in O_{\text{ref}} \mid o_i \notin O_{\text{end}} \vee \text{IoU}\left(m_i^{\text{ref}}, m_i^{\text{end}}\right) < \tau \quad (2)$$

where m_i^{ref} and m_i^{end} are the masks of object o_i in the designated first and last frames, respectively, and τ is the IoU threshold.

2) Bidirectional Cross Validation Algorithm: Step 1: Single-Sequence Change Candidate Extraction

For each sequence (ascending S_{asc} and descending S_{desc}).

1) *First-Last Frame Object Matching:* Extract reference objects $\{(id_i, m_i^{\text{ref}})\}$ from a designated starting frame $F^{(k)}$ (typically $k = 0$ or $k = N$), then retrieve their corresponding masks m_i^{end} in the terminal frame $F^{(N+1)}$.
2) *IoU Calculation and Change Flagging:* For each object id_i, flag it as a change candidate if either condition is met:
   a) *Condition 1:* id_i is absent in the last frame (ID loss).
   b) *Condition 2:* $\text{IoU}(m_i^{\text{ref}}, m_i^{\text{end}}) < \tau$.
3) *Candidate Mask Generation:* Merge masks of all flagged objects to generate candidate masks C_{asc} and C_{desc}.

Mathematical Expression:

$$C_{\text{asc}} = \bigcup \left\{ m_i^{\text{ref}} \mid id_i \notin O_{\text{end}}^{\text{asc}} \vee \text{IoU}\left(m_i^{\text{ref}}, m_i^{\text{end}}\right) < \tau \right\}. \quad (3)$$

Step 2: Cross-Sequence Pseudo-Change Filtering

1) *Cross-Sequence Candidate Comparison:* Treat C_{asc} and C_{desc} as two independent sets. Iterate over all candidate pairs ($m_A \in C_{\text{asc}}$, $m_B \in C_{\text{desc}}$).
2) *High-IoU Elimination:* If $\text{IoU}(m_A, m_B) > \tau_{\text{cross}}$, m_A and m_B are considered pseudo-changes (e.g., caused by bidirectional tracking failures due to environmental noise) and removed.
3) *Final Change Mask Generation:* Merge remaining candidates

$$M_{\text{diff}} = (C_{\text{asc}} \setminus \{m_A \mid \exists m_B \in C_{\text{desc}}, \text{IoU}(m_A, m_B) > \tau_{\text{cross}}\}) \cup (C_{\text{desc}} \setminus \{m_B \mid \exists m_A \in C_{\text{asc}}, \text{IoU}(m_A, m_B) > \tau_{\text{cross}}\}). \quad (4)$$

IV. Experiments

A. Data Description

Three representative publicly available change detection datasets were selected to validate the performance of BiSAM-CD.

WHU-CD [46] is a building change detection dataset based on postearthquake reconstruction in Christchurch, New Zealand, in 2012 and 2016. It contains an aerial remote sensing image pair with a resolution of 32 507 × 15 354, covering an area of 20.5 km^2. To ensure fairness for testing with 1024 × 1024 image inputs, the source image was cropped into 31 744 × 14 336 from left to right and top to bottom. Subsequently, it was further cropped into 347/43/44 and 5552/688/704 patches with resolutions of 1024 × 1024 and 256 × 256, respectively, for training, validation, and testing.

LEVIR-CD [16] is a large-scale remote sensing building change detection dataset comprising 637 image pairs from Google Earth with a resolution of 1024 × 1024, spanning 5–14 years.

SECOND [47] dataset consists of 4662 pairs of 512 × 512 aerial images with a spatial resolution ranging from 0.5 to 3 m per pixel, annotated for semantic change detection (SCD) in urban environments across cities such as Hangzhou, Chengdu, and Shanghai. It focuses on six fundamental land cover categories: nonvegetated ground surfaces, trees, low vegetation, water, buildings, and playgrounds, designed to encapsulate diverse geographical transitions. To ensure precise change localization, the dataset provides land-cover map pairs alongside nonchange masks, which explicitly differentiate altered pixels from unchanged regions within the same semantic class.

B. Experimental Setup

The proposed *BiSAM-CD* is implemented using the PyTorch framework and runs on NVIDIA RTX 4090D GPUs.

1) Evaluation Metrics: To objectively evaluate the performance of BiSAM-CD, five key evaluation metrics are adopted: precision (Pre), recall (Rec), $F1$ Score ($F1$), intersection over union (IoU), and overall accuracy (OA). Pre measures the proportion of actual positive samples among those predicted as positive by the model; Rec measures the proportion of correctly predicted positive samples among actual positives; $F1$ is the harmonic mean of precision and recall; IoU measures the overlap between predicted results and ground truth labels; OA represents the proportion of correctly predicted samples

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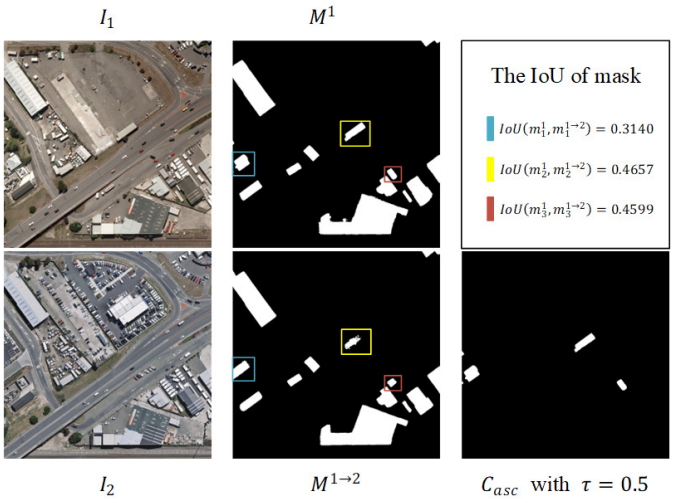


Fig. 3. Example for single-sequence change candidate extraction. I_1 and I_2 represent two-temporal remote sensing images. M^1 denotes the mask at the initial moment, and $M^{1\to2}$ is the tracking result of the object in I_1 within I_2. The right-hand side annotates the IoU values of different mask regions. C_{asc} is the finally extracted change-candidate region, demonstrating the change-detection result based on $\tau = 0.5$.

among the total samples. The formulas for each metric are as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$F1 = \frac{2}{\text{Rec}^{-1} + \text{Pre}^{-1}} \tag{7}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{8}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{9}$$

where true positive (TP) represents true positives, false positive (FP) represents false positives, FN (false negative) represents false negatives, and TN (true negative) represents true negatives. Among the five metrics, the $F1$ score considers both precision and recall, while IoU evaluates the overlap between predicted and ground truth regions. These metrics are commonly used to assess overall model performance and better reflect its generalization capability.

2) Prompt Generation: Object prompts (points, boxes, and masks) are generated from ground-truth labels: points are derived from mask centroids, boxes from minimum bounding rectangles, and masks directly from binary annotations.

The key parameters are set as IoU threshold $\tau = 0.5$ and interpolated frames $N = 1$. Four SAM2 checkpoints (Tiny, Small, Base+, and Large) are evaluated to assess scalability.

C. Quantitative and Qualitative Results

To validate the effectiveness of BiSAM-CD in change detection tasks, the article compares it with several representative methods from recent years, including supervised methods (FC-EF [12], ChangeFormer [14], TinyCD [48], RS-Mamba [49], and SAM-CD [50]), and SS methods (SemiCD [51], UniMatch [52], C2F-SemiCD [24], and Unimatch V2 [53]).

TABLE I

COMPARISON OF SUPERVISED (S), SS, AND ZS METHODS ON WHU-CD DATASET. METRICS (PRECISION, RECALL, $F1$, IOU, AND OA) ARE REPORTED IN PERCENTAGES (%). FOR BISAM-CD, PARAMETERS ARE SET AS $\tau = 0.5, N = 1$, AND MASK PROMPTS ARE DERIVED FROM WHU DATASET

Models	Type	WHU-CD				
		Pre.	Rec.	F1	IoU	OA
FC-Siam-Conc$_{18}$	S	86.31	91.77	89.53	80.27	99.09
ChangeFormer$_{22}$	S	93.67	87.01	90.24	83.60	99.26
TinyCD$_{23}$	S	92.62	91.85	90.85	85.69	99.40
RS-Mamba$_{24}$	S	93.55	91.36	91.78	87.53	99.43
SAM-CD$_{24}$	S	92.12	92.81	91.02	87.58	99.46
SemiCD$_{22}$	SS	88.12	85.91	87.00	78.19	99.02
UniMatch$_{23}$	SS	84.58	90.66	87.51	79.02	99.02
C2F-SemiCD$_{24}$	SS	90.03	86.63	88.31	80.32	99.03
UnimatchV2$_{25}$	SS	90.48	91.14	89.89	81.17	99.13
Ours (tiny)	ZS	**86.58**	87.53	84.49	79.85	99.20
Ours (small)	ZS	84.49	85.39	82.67	77.72	99.14
Ours (base+)	ZS	84.10	**89.97**	**85.31**	**80.66**	**99.38**
Ours (large)	ZS	83.64	86.87	82.68	77.70	99.19

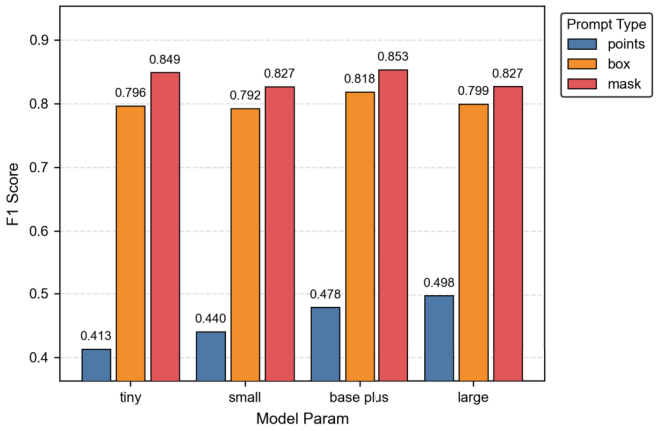


Fig. 4. Impact of prompt types (points/boxes/masks) on BiSAM-CD performance. Parameters are set as $\tau = 0.5$ and $N = 1$ on WHU dataset.

Quantitative comparisons with the above methods (see Table I) reveal that our ZS approach achieves competitive performance. Using the Base+ checkpoint, our method attains $F1 = 85.31\%$ and IoU = 80.66%, outperforming some SS models like UniMatch (IoU = 79.02%) and approaching supervised methods such as FC-Siam-Conc (IoU = 80.27%). Larger SAM2 variants exhibit a precision-recall tradeoff, with the Large checkpoint showing lower precision (83.64%) but higher recall (86.87%). The superiority of mask prompts is evident in Fig. 4, where mask-driven results (Base+: $F1 = 85.3\%$) surpass box (81.8%) and point (47.8%) prompts, underscoring the importance of shape priors. Tiny and Small checkpoints demonstrate robustness, with $F1$ drops below 5% compared to Base+, highlighting SAM2's scalability.

Visual comparisons (see Fig. 5) reveal distinct behavioral patterns across prompt types: for contiguous building clusters Fig. 5(a), point and box prompts (BiSAM-p/b) maintain

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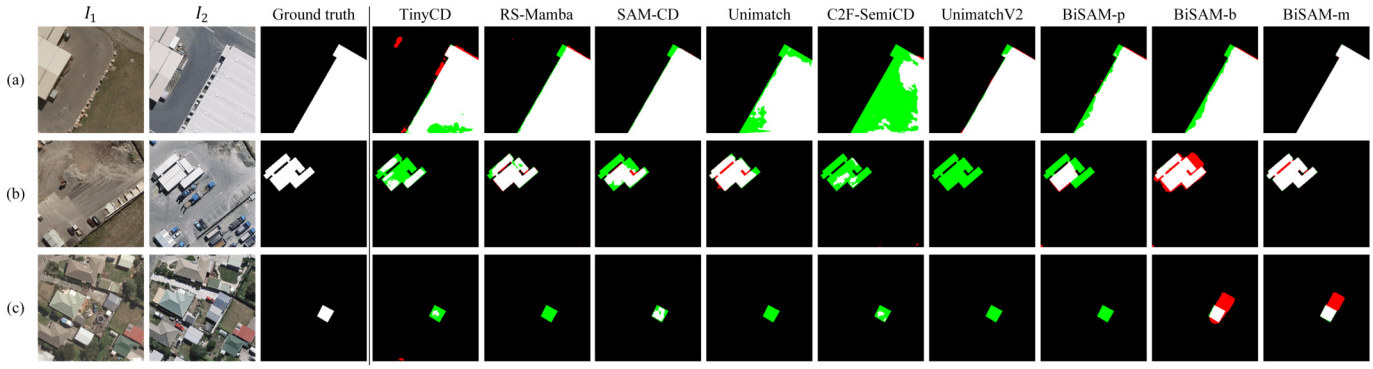


Fig. 5. Qualitative comparison between BiSAM-CD with different prompt types (point/box/mask) and ground truth on the WHU-CD dataset. White indicates TPs, black indicates TNs, red denotes FPs, and green denotes FNs. (a) Large building example. (b) Complex-shaped example. (c) Geometric distortion example.

complete building outlines but exhibit edge fragmentation due to interference from adjacent nontarget features, whereas mask prompts (BiSAM-m) produce coherent boundaries aligning closely with ground truth. In complex-shaped structures Fig. 5(b), BiSAM-p suffers from partial target capture as single-point initialization fails to propagate through entire buildings, while BiSAM-b introduces FPs by incorporating shadowed areas within bounding boxes; BiSAM-m demonstrates superior shape fidelity with minimal artifacts. Cases with intertemporal geometric distortions expose inherent limitations: BiSAM-p misses spatially shifted targets entirely, whereas both BiSAM-b and BiSAM-m not only correctly identify displaced buildings but also generate collateral false detections on neighboring unchanged structures due to positional ambiguities.

While Table I demonstrated BiSAM-CD's upper-bound performance using ground truth masks, practical deployments often lack such annotations. To rigorously evaluate BiSAM-CD's adaptability to diverse prompt sources in real-world scenarios where ground truth masks are unavailable, we conducted comprehensive experiments on WHU-CD and LEVIR-CD datasets using three types of automatically generated prompts. This experiment systematically evaluates the framework's robustness when initialized with: 1) OWLv2 [8] (base/large) as representative generic object detectors; 2) SAMPoly [11] as a domain-specialized building detector; and 3) SAM2's automatic masks.

For WHU-CD and LEVIR-CD, we generated: *Box prompts*: OWLv2-B/L (B uses "single building" and L uses "roof" as text prompts), SAMPoly (default settings). *Mask prompts*: SAM2 with default settings.

Quantitative evaluation (see Table II) shows SAM2's auto-masks achieved 61.78% recall but only 9.04% precision ($F1$ = 14.02%) on WHU-CD, while SAMPoly yielded the highest $F1$-score (57.43%) among automated prompts on LEVIR-CD. Qualitative analysis (see Figs. 6 and 7) reveals that for large buildings [see Figs. 6(a) and 7(a)], OWLv2-L generates more complete bounding boxes compared to OWLv2-B, resulting in better-preserved structures in the change masks, while SAMPoly's domain-specific design produces the most accurate building contours. In cases of complex shapes [see Figs. 6(b) and 7(b)], SAM2's auto-masking consistently incorporates nontarget objects such as vehicles and shadows, leading to noticeable FPs. Regarding geometric distortions [see Fig. 6(c)], all prompt types demonstrate alignment-sensitive failures, with OWLv2-B exhibiting more severe fragmentation in tracked objects.

Despite poor single-class performance, SAM2 auto-masks exhibited a unique capability to capture diverse objects. We hypothesized this property might benefit multiclass change detection, where exhaustive class-specific prompts are impractical.

Using SAM2 auto-masks as sole prompts, we compared against supervised semantic change detectors (HRSCD [54], ChangeMask [55], SSCD [56], BiSRNet [56], TED [57], SMNet [58], ScanNet [57], and MambaSCD [59]) under standard $F1$, mIoU, and OA metrics.

Quantitative evaluation (see Table III) shows BiSAM-CD achieves 39.35% $F1$ without class-specific tuning. Outperforms HRSCD-S1 in mIoU (50.76% versus 47.57%). The qualitative results (see Fig. 8) demonstrate that SAM2 successfully identifies major change regions in multiobject scenarios (e.g., newly constructed buildings and expanded roads) on the SECOND dataset, but exhibits typical error patterns including: 1) false detections of ground stains or temporary debris as meaningful changes [see Fig. 8(a) and (e)]; 2) misclassification of vegetation texture variations caused by seasonal or illumination changes [see Fig. 8(b)]; and 3) erroneous detection of nonstructural building modifications such as roof recoloring or decorative alterations [see Fig. 8(c) and (d)].

D. Ablation Studies

Table IV validates the necessity of each module through controlled experiments. The baseline (raw SAM2) achieves limited performance ($F1$ = 54.66%), highlighting the challenge of direct temporal comparison without temporal consistency mechanisms. Introducing our pipeline (without Modules A/B) slightly improves $F1$ to 57.23%, suggesting that bidirectional sequence generation alone offers marginal benefits.

The critical role of Module A (TOTC-CSCV) is evident: its integration boosts $F1$ by 21.16%–75.82%, primarily by suppressing pseudo-changes through cross-sequence validation (see Fig. 9). Surprisingly, Module B (BFSCAI) alone degrades

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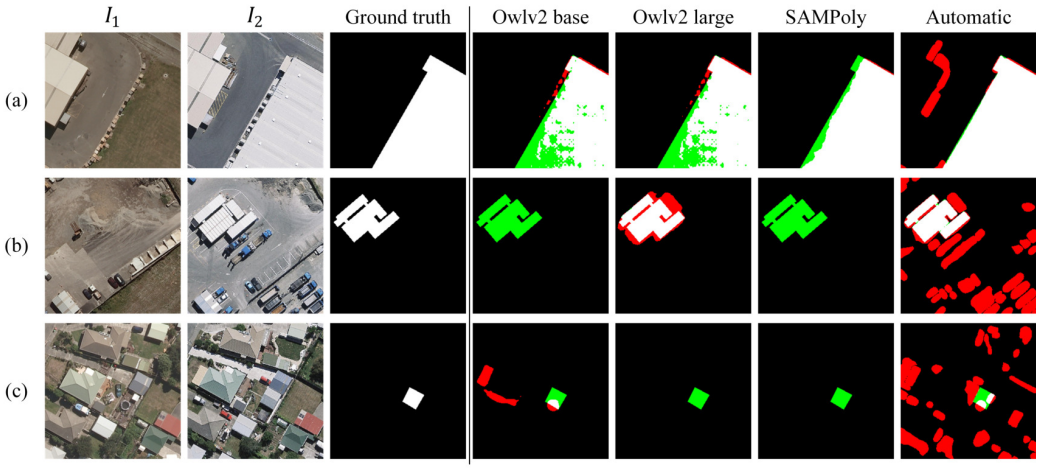


Fig. 6. Qualitative comparison of BiSAM-CD with different automated prompts on WHU-CD dataset. White indicates TPs, black indicates TNs, red denotes FPs, and green denotes FNs. (a) Large building example. (b) Complex-shaped example. (c) Geometric distortion example.

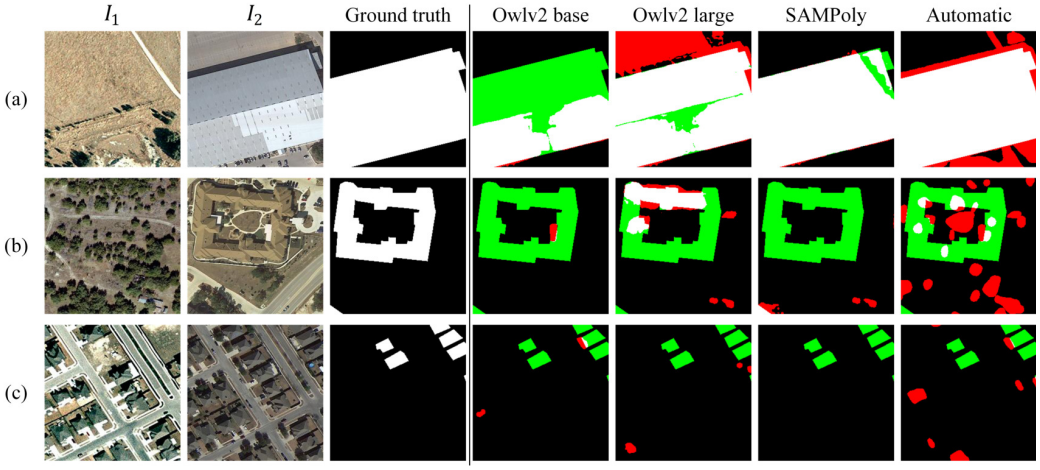


Fig. 7. Qualitative comparison of BiSAM-CD with different automated prompts on LEVIR-CD dataset. White indicates TPs, black indicates TNs, red denotes FPs, and green denotes FNs. (a) Large building example. (b) Complex-shaped example. (c) Residential building complex example.

performance ($F1 = 48.71\%$ versus baseline 54.66%), revealing that adaptive interpolation exacerbates tracking errors when pseudo-changes remain unfiltered. This counterintuitive result underscores a key insight: SAM2's temporal memory is vulnerable to interpolated noise without Module A's corrective mechanism.

Only when combined do both modules exhibit synergy: Module B stabilizes long-term tracking (see Fig. 10) while Module A ensures mask consistency, jointly achieving the best $F1$ (81.83%) with a 6.01% gain over Module A alone. This demonstrates that BFSCAI's value is conditional on TOTC-CSCV's prior noise suppression.

E. Parameter Sensitivity and Reference Frame Robustness Validation

To comprehensively evaluate parameter interactions, we construct a full-factorial experiment with.

1) *SAM2 Variants:* Tiny, Small, Base+, and Large.
2) *Interpolation Frames:* $N \in 1, 2, 3$.
3) *IoU Thresholds:* $\tau \in 0.3, 0.4, 0.5, 0.6, 0.7$.
4) *Reference Frames:* First, penultimate.

Parameter sensitivity analysis [see Fig. 11(a) and (b)] reveals optimal configurations. A value of 0.5 optimally balances precision and recall ($F1 = 81.83\%$), while higher thresholds (e.g., 0.7) preferentially suppress TPs, indicating SAM2's mask propagation tends to preserve object shapes but requires moderate tolerance for geometric distortions. The stability of $F1$ scores within the range [0.4, 0.6] demonstrates the robustness of our pseudo-change filtering strategy to the IoU threshold. The superiority of $N = 1$ [Fig. 11(b)] demonstrates that a single interpolated frame sufficiently mitigates appearance gaps for abrupt building changes (e.g., demolition/construction in WHU-CD), whereas excessive interpolation ($N \geq 2$) introduces synthetic artifacts that disrupt SAM2's temporal memory. This phenomenon aligns with the binary transition property of urban changes, where intermediate states (e.g., partially collapsed roofs) are particularly rare in bi-temporal satellite observations with long-term intervals.

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TABLE II

PERFORMANCE COMPARISON OF BISAM-CD WITH AUTOMATICALLY GENERATED PROMPTS VERSUS SUPERVISED (S) AND SS BASELINES ON WHU-CD AND LEVIR-CD DATASETS. METRICS (PRECISION, RECALL, $F1$, IOU, AND OA) ARE REPORTED IN PERCENTAGES (%). FOR BISAM-CD, PARAMETERS ARE SET AS $\tau = 0.5$ AND $N = 1$, WHERE BOX PROMPTS ARE GENERATED BY OWLV2-B/L AND SAMPOLY, WHILE MASK PROMPTS ARE DERIVED FROM SAM2'S AUTO-GENERATED SEGMENTATION MASKS

Models	Type	WHU-CD					LEVIR-CD				
		Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
FC-Siam-Conc$_{18}$	S	86.31	91.77	89.53	80.27	99.09	90.76	88.68	89.63	81.36	98.96
ChangeFormer$_{22}$		93.67	87.01	90.24	83.60	99.26	91.10	88.72	89.81	81.63	98.97
TinyCD$_{23}$		92.62	91.85	90.85	85.69	99.40	92.30	89.88	90.36	83.56	99.09
RS-Mamba$_{24}$		93.55	91.36	91.78	87.53	99.43	91.40	88.42	89.88	81.65	99.07
SAM-CD$_{24}$		92.12	92.81	91.02	87.58	99.46	91.21	90.70	90.42	84.09	99.14
SemiCD$_{22}$	SS	88.12	85.91	87.00	78.19	99.02	89.26	81.33	85.12	73.96	98.55
UniMatch$_{23}$		84.58	90.66	87.51	79.02	99.02	93.39	80.13	86.22	75.68	98.70
C2F-SemiCD$_{24}$		90.03	86.63	88.31	80.32	99.03	86.32	84.09	84.71	81.29	99.01
UnimatchV2$_{25}$		90.48	91.14	89.89	81.17	99.13	88.81	86.58	87.20	82.43	99.04
Ours (owlv2-B)	ZS	37.15	45.23	37.72	30.67	96.86	45.44	39.75	35.83	26.80	94.27
Ours (owlv2-L)		42.55	55.36	43.88	36.23	96.99	48.11	52.50	46.85	33.66	94.91
Ours (sampoly)		74.61	68.26	65.61	59.13	98.56	68.37	56.49	57.43	44.80	96.68
Ours (automatic)		9.04	61.78	14.02	8.70	78.96	20.45	56.19	26.55	16.55	88.24

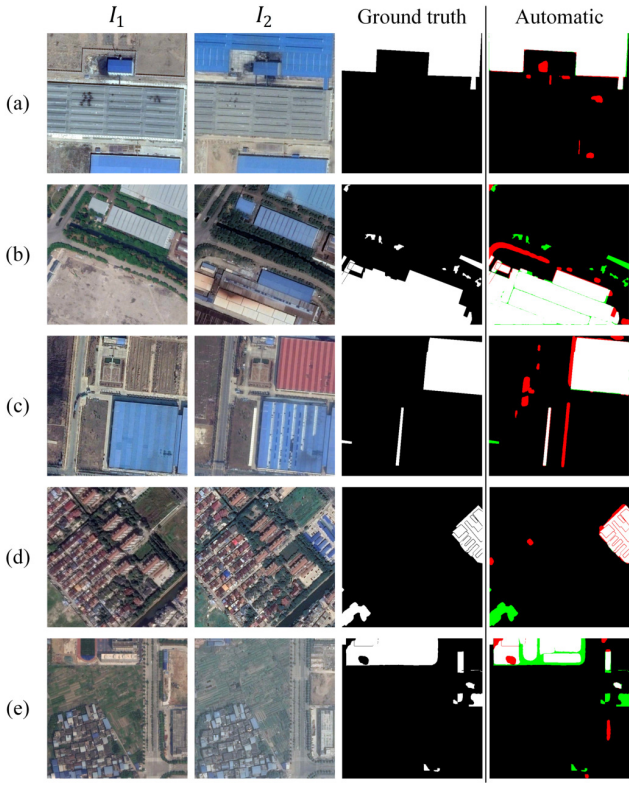


Fig. 8. Qualitative comparison of BiSAM-CD with SAM2 auto-masks on the SECOND dataset. White indicates TPs, black indicates TNs, red denotes FPs, and green denotes FNs. (a)–(c) Industrial area examples. (d) and (e) Urban area examples.

Penultimate-frame matching narrows the interquartile range (IQR) versus the first-frame [see Fig. 12(b) versus (a)], demonstrating enhanced stability against tracking drift. Minimum $F1$ increases in Tiny, Small and Base+ models, proving that reducing the temporal gap mitigates catastrophic tracking failures.

TABLE III

PERFORMANCE COMPARISON OF BISAM-CD WITH AUTOMATICALLY GENERATED PROMPTS VERSUS SUPERVISED (S) BASELINES ON SECOND DATASET. METRICS ($F1$, MIOU, AND OA) ARE REPORTED IN PERCENTAGES (%). FOR BISAM-CD, PARAMETERS ARE SET AS $\tau = 0.5$ AND $N = 1$, WHERE THE MASK PROMPTS WERE AUTOMATICALLY GENERATED BY SAM2'S SEGMENTATION MASKS

Models	Type	SECOND		
		F1	mIoU	OA
HRSCD-S1$_{19}$	S	33.04	47.57	62.56
HRSCD-S2$_{19}$		39.79	63.80	82.93
HRSCD-S3$_{19}$		41.78	64.96	91.17
HRSCD-S4$_{19}$		47.68	70.58	83.88
ChangeMask$_{22}$		50.54	71.39	85.53
SSCD$_{22}$		82.80	72.14	85.62
BiSRNet$_{22}$		53.59	72.69	85.80
TED$_{24}$		52.61	72.19	85.50
SMNet$_{23}$		50.48	71.62	84.29
ScanNet$_{24}$		55.21	72.48	86.13
MambaSCD-T$_{24}$		54.51	73.33	90.44
MambaSCD-S$_{24}$		55.72	73.41	86.48
MambaSCD-L$_{24}$		55.32	73.68	90.36
Ours	ZS	**39.35**	**50.76**	**78.43**

F. Computational Efficiency Analysis

Our experiments demonstrate BiSAM-CD's practical efficiency through three key findings (see Table V).

1) Under box prompts with single mid-frame interpolation, the framework processes 1024 × 1024 images with near-real-time latency (0.0325 s for Base+) and sub-GB memory footprint (967 MB), outperforming supervised counterparts like ChangeFormer (0.1277 s, 19.2 GB) and SAM-CD (0.1306 s, 1.84 GB) by 4× speedup.

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TABLE IV

ABLATION STUDY OF MODULE CONTRIBUTIONS ON WHU-CD DATASET. A: TEMPORAL OBJECT TRACKING CONSISTENCY AND CROSS-SEQUENCE CHANGE VERIFICATION (TOTC-CSCV). B: BFSCAI. BF: BIDIRECTIONAL FRAMEWORK. OURS: SAM2 (BASE+) WITH BOX PROMPTS, $\tau = 0.5$, AND $N = 1$

Models	A	B	WHU-CD				
			Pre.	Rec.	F1	IoU	OA
Baseline	×	×	49.05	78.07	54.66	47.78	97.29
+BF	×	×	52.16	75.65	57.23	50.45	98.38
+BF+A	✓	×	72.40	87.14	75.82	69.34	99.03
+BF+B	×	✓	46.08	61.84	48.71	42.20	97.92
Ours	✓	✓	**79.22**	**88.81**	**81.83**	**75.62**	**99.04**

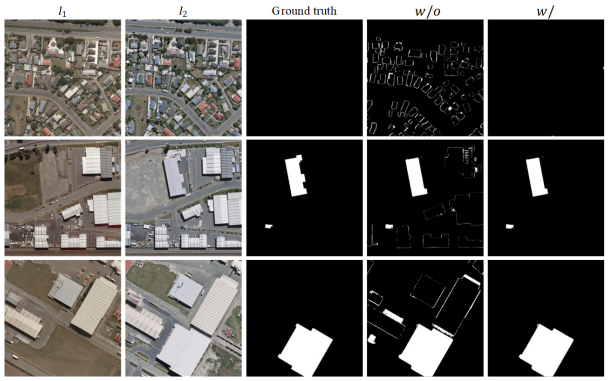


Fig. 9. Visual comparison of pseudo-change suppression via the TOTC-CSCV. Column 1: earlier temporal image I_1. Column 2: later temporal image I_2. Column 3: ground truth. Column 4: detection results without TOTC-CSCV (w/o). Column 5: detection results with TOTC-CSCV ($w/$).

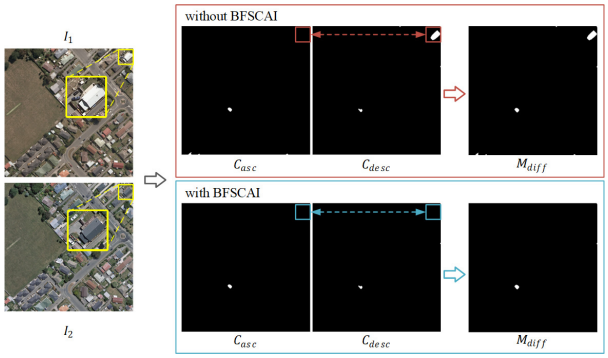


Fig. 10. Bidirectional tracking consistency improvement via BFSCAI. (Left) Bi-temporal images I_1 (top) and I_2 (bottom). (Right) Results without BFSCAI (top). (Right) Results with BFSCAI (bottom).

2) The system scales linearly with object complexity (see Fig. 13), with worst case 100-object scenarios maintaining reasonable resource demands (1.79-GB peak memory for Base+).
3) Model variants provide flexible tradeoffs: tiny achieves edge-device compatibility (0.022 s, 698 MB) while large retains workstation-level efficiency (0.0523 s, 1.84 GB).

V. DISCUSSION

A. Practical Implications and Tradeoffs

The experimental results demonstrate that mask-based prompts achieve the highest accuracy ($F1$ = 85.31% for Base+), aligning with SAM2's design for shape-aware segmentation. However, acquiring pixel-level masks for real-world applications remains labor-intensive, especially in large-scale monitoring scenarios. In contrast, box prompts offer a pragmatic compromise ($F1$ = 81.83%) between annotation effort and detection accuracy, making them preferable for time-sensitive tasks like disaster response, where rapid deployment outweighs marginal accuracy gains.

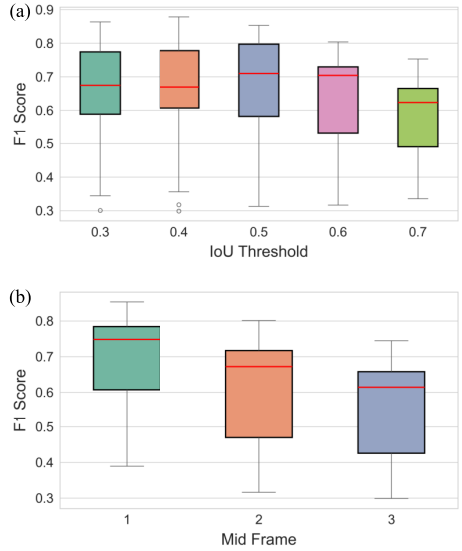


Fig. 11. Parameter sensitivity analysis of BiSAM-CD. (a) Sensitivity of $F1$ score to IoU thresholds. (b) Effect of mid-frame interpolation counts on $F1$ score.

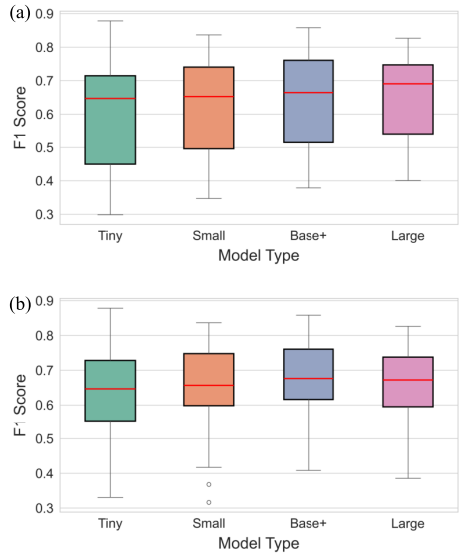


Fig. 12. Impact of reference frame selection on tracking robustness. (a) First-versus-last frame matching. (b) Penultimate-versus-last frame matching.

B. Dependence on External Detection Models

While Section IV-C demonstrates BiSAM-CD's compatibility with external detectors (e.g., OWLv2 and SAMPoly), the framework fundamentally operates as a ZS system. The dependency on prompts is a deliberate design choice to balance flexibility and performance.

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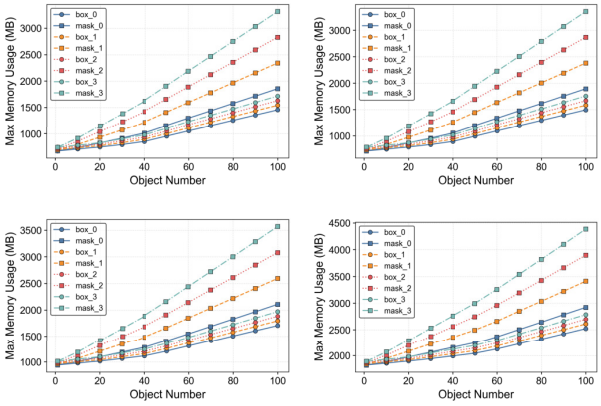


Fig. 13. Memory usage patterns of BiSAM-CD variants across different operational parameters (Tiny, Small, Base+, and Large), prompt types (box and mask), and intermediate frames (0, 1, 2, and 3). All measurements were conducted on 1024 × 1024 images with NVIDIA RTX 4090D GPUs.

Auto-mask mode (zero manual input) achieves 61.78% recall on WHU-CD (see Table II).

Manual prompts (boxes/masks) boost precision to 85.31% $F1$.

This dichotomy mirrors foundational models like SAM, where prompt engineering enhances—rather than defines—ZS ability. Quantitative evidence in Tables II and III confirm that prompt quality affects performance bounds but does not invalidate the training-free paradigm, as auto-masks alone outperform some supervised baselines. Furthermore, auto-mask mode prioritizes recall for rapid screening, while manual prompts enhance precision for accuracy-critical tasks. This flexibility ensures deployability across scenarios—from disaster response (recall-driven) to urban planning (precision-driven).

C. Strengths and Limitations of SAM2

The success of our method hinges on SAM2's ZS generalization and temporal memory mechanism, which enable robust tracking across interpolated sequences. However, SAM2 exhibits inherent limitations in remote sensing contexts.

Sensitivity to Seasonal Variations: While Module B mitigates illumination changes through adaptive interpolation, long-term seasonal shifts still cause tracking failures, as SAM2 lacks domain-specific photometric adaptation.

Scale Ambiguity: High-resolution images occasionally lead to object fragmentation, triggering erroneous ID assignments in the TOTC-CSCV module.

D. Design Choices and Tradeoffs in a Training-Free Paradigm

The design of the adaptive interpolation module [BFSCAI, (1)] was guided by the overarching constraint of maintaining a fully training-free paradigm. While more sophisticated photometric alignment techniques exist, such as optical flow warping [29] or deep learning-based illumination normalization, they typically require pretrained models or iterative optimization on the image pair itself. These approaches would introduce external dependencies, computational overhead, and potential task-specific tuning, thereby contradicting our core objective of ZS, immediate deployment. Our heuristic approach strikes a deliberate balance: it is sufficient to mitigate dominant appearance variations (e.g., illumination shifts) and unlock SAM2's temporal memory capability, as evidenced by the performance gain in ablation studies (see Table IV), without compromising the framework's generality and efficiency. This tradeoff underscores a key insight for training-free systems: practical effectiveness often outweighs theoretical optimality.

TABLE V

COMPUTATIONAL EFFICIENCY COMPARISON ON 1024 × 1024 IMAGES. VALUES IN PARENTHESES INDICATE RESOURCE CONSUMPTION WHEN PROCESSING 100 OBJECTS. ALL COMPETING METHODS ARE EVALUATED AT THEIR NATIVE INPUT RESOLUTIONS (256 × 256 UNLESS NOTED) WITH BATCH SIZES ADJUSTED TO MATCH 1024 × 1024 PIXEL COVERAGE. "TYPE" DENOTES SUPERVISED (S), SS, AND ZS APPROACHES. OURS: WITH BOX PROMPTS, $\tau = 0.5$, AND $N = 1$

Models	Type	Size	Time ($s \cdot 10^{-2}$)	Memory (MB)
ChangeFormer$_{22}$	S	256	12.77	19230.85
TinyCD$_{23}$		256	4.02	866.46
RS-Mamba$_{24}$		256	16.13	2259.71
SAM-CD$_{24}$		1024	13.06	1841.06
SemiCD$_{22}$	SS	256	7.70	10420.73
UniMatch$_{23}$		256	3.79	1230.09
C2F-SemiCD$_{24}$		256	15.45	2934.41
UniMatch V2$_{25}$		252	12.71	2792.30
Ours (tiny)	ZS	1024	2.20 (186.36)	697.70 (1541.67)
Ours (small)			2.68 (180.39)	737.70 (1582.61)
Ours (base+)			3.25 (181.61)	967.39 (1793.93)
Ours (large)			5.23 (188.51)	1844.15 (2610.24)

E. Parameter-Driven Insights

Our sensitivity analysis reveals two deployment guidelines as follows.

Efficiency-Accuracy Tradeoff: For latency-critical scenarios, the Tiny variant achieves an $F1$ score exceeding 80% with an inference time of 0.022 s (see Table V), under the configuration of $\tau = 0.5$ and one interpolated frame ($N = 1$). In contrast, the Base+ model attains the highest $F1$ score (85.31%) for accuracy-critical applications.

Temporal Robustness: As shown in Fig. 12, penultimate-frame referencing notably narrows the IQR of $F1$ scores for Tiny/Small models, indicating that shorter temporal gaps help alleviate SAM2's tracking drift over time.

Interpolation-Memory Synergy: The necessity of coupling BFSCAI with TOTC-CSCV (see Table IV) underscores a fundamental principle: SAM2's temporal memory requires pseudo-change filtering (Module A) before interpolation (Module B) can enhance robustness. This explains why Module B alone degrades performance—unfiltered environmental noise (e.g., shadows and seasonal variations) propagates through interpolated frames. Future work may explore dynamic frame scheduling (e.g., adaptive N based on scene volatility) to further optimize this synergy.

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F. Toward Fully Autonomous Change Detection

The current pipeline achieves annotation-free detection but still necessitates manual prompt initialization (e.g., selecting detection models or text queries). A promising direction is to unify prompt generation and change analysis within a single framework. For instance, combining SAM2 with self-supervised feature clustering could automatically identify candidate changed regions, which are then refined via bidirectional tracking. Exploratory experiments on the SECOND dataset (see Table III) reveal BiSAM-CD's unique potential in ZS multiclass SCD: Using SAM2's auto-generated prompts, it achieves 39.35% $F1$-score, outperforming some supervised models. As shown in Fig. 8, the framework successfully captures cross-category changes but struggles with nonstructural alterations. This suggests that class-aware prompt generation via self-supervised clustering could bridge the gap toward fully autonomous multiclass detection, while mitigating SAM2's sensitivity to texture artifacts.

Unlike prior SAM adaptations limited to architectural tweaks or single-temporal analysis, our framework fundamentally exploits SAM2's temporal memory for cross-temporal reasoning, establishing a new paradigm for foundation-model-based change detection. This shift from per-frame segmentation comparison to bidirectional object persistence validation addresses the core challenges of illumination variance and seasonal shifts, paving the way for fully autonomous systems.

G. Broader Impact and Applications

Beyond urban monitoring, our method could revolutionize environmental conservation (e.g., deforestation tracking) and infrastructure assessment (e.g., postdisaster damage mapping) by enabling rapid, low-cost change detection. The training-free paradigm is particularly valuable for resource-limited regions, where annotated data and computational resources are scarce.

VI. Conclusion

This article presents a novel training-free framework for remote sensing change detection by leveraging the temporal memory mechanism of SAM2. Through bidirectional sequence construction and cross-temporal consistency validation, our method achieves competitive accuracy on the WHU-CD dataset without requiring annotated training data. The framework demonstrates consistent performance across diverse scenarios: urban building changes (WHU-CD: 85.31% $F1$), long-term land cover monitoring (LEVIR-CD: 57.43% $F1$), and multiclass semantic transitions (SECOND: 39.35% $F1$), establishing its generalization capability. Key innovations include.

Bidirectional Tracking Strategy: Adaptive interpolation and penultimate-frame referencing significantly reduce tracking failures (see Section IV-E), effectively handling illumination/seasonal variations in LEVIR-CD's 5–14 year gaps.

Lightweight Optimization Modules: TOTC-CSCV (Module A) effectively eliminates pseudo-changes through cross-sequence IoU validation, while BFSCAI (Module B) ensures object continuity despite imperfections in OWLv2's prompts (see Figs. 6 and 7).

ZS Adaptability: The pipeline achieves near-real-time processing (0.0325 s/image for Base+ at 1024 $\times$ 1024) with sub-GB memory (see Table V), outperforming supervised models (e.g., ChangeFormer) by 4$\times$ speedup.

Despite its promise, the method currently relies on external detection models for prompt initialization, and its performance on long-term seasonal changes remains limited. For binary change detection (WHU-CD), SAM2's auto-masks achieve high recall (61.78%) but low precision (9.04%), indicating a need for improved FP suppression. For multiclass detection (SECOND), the framework achieves 39.35% $F1$ without class-specific tuning, but struggles with semantic ambiguity [e.g., misclassifying vegetation texture variations as changes, Fig. 8(b)]. Future work will focus on: 1) domain-specific memory augmentation to address long-term seasonal changes; 2) unified pipelines for binary and multiclass change detection; and 3) exploring prompt clustering as a potential strategy. This research advances the practical deployment of foundation models in resource-constrained scenarios, offering a scalable solution for rapid environmental monitoring.

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